

How to Design a Simple Multi-Timeframe Trend Strategy on Bitcoin

Authors

Radovan Vojtko*

David Mesíček†

* CEO & Head of Research, Quantpedia

† Quant Analyst, Quantpedia. Corresponding author: david.mesicek@quantpedia.com

Abstract

This study investigates the construction and evaluation of a simple multi-timeframe trend-following strategy applied to Bitcoin. We begin with a naïve baseline model and progressively enhance it using daily–hourly trend confirmation and a trailing exit mechanism. Using BTC/USD data from Gemini Exchange between 2018 and 2025, we demonstrate how multi-timeframe structure improves stability, reduces drawdowns, and generates superior risk-adjusted performance compared to both buy-and-hold and a pure MACD strategy. Results show meaningful improvements in Sharpe and Calmar ratios as complexity increases logically without parameter optimization.

1 Introduction

Bitcoin has emerged as one of the most actively traded global assets, characterized by structural upward drift, deep cyclical drawdowns, and high intraday volatility. These features make it a natural candidate for trend-following research. Technical analysis remains a dominant approach among both retail and institutional traders, especially in markets where reliable fundamentals are limited.

Trend-following strategies based on moving averages or oscillators attempt to capture persistent directional movement. Bitcoin's strong trending periods combined with prolonged consolidations make it ideally suited for multi-timeframe analysis, where higher timeframe structure reduces noise and improves signal quality on execution timeframes.

The aim of this study is to demonstrate a step-by-step, logically constructed Bitcoin trend strategy that evolves from a basic MACD signal to a multi-timeframe model with improved exits. Because Bitcoin exhibits long-term appreciation and large upward cycles, we restrict our study to long-only systems. This approach is consistent with standard crypto research methodology and is easier to implement in practice.

2 Background

Trend-following systems attempt to exploit autocorrelation in financial returns by entering positions when momentum is positive and exiting when trend conditions deteriorate. Traditional moving-average systems, MACD oscillators, and price-action filters attempt to capture this effect on a single timeframe.

However, single-timeframe trend strategies often suffer from noise and whipsaws, particularly in high-volatility assets like Bitcoin. Multi-timeframe trend confirmation using a slow trend on the daily chart and executing on faster charts offers a stable and widely used solution, reducing false signals and aligning trade direction with dominant market structure.

Our research follows the principle of **structural robustness over parameter tuning**:

1. Establish a simple baseline.
2. Add higher timeframe confirmation.
3. Improve exit efficiency with trailing logic.

This methodology is aligned with industry best practices for systematic strategy development.

3 Methodology

3.1 Baseline: MACD Pure (H1)

- **Indicator:** MACD(12, 26, 9) on 1H candles
- **Condition:** Long-only
- **Entry:** MACD line crosses above signal line
- **Exit:** MACD line crosses below signal line
- **Execution:** On close of the 1H bar
- **Rationale:** Establish a basic, non-optimized momentum model

3.2 Multi-Timeframe Confirmation: D1–H1 MACD

Daily trend defines directional bias. Hourly MACD manages entries.

- **Daily Trend Filter:** Bullish when Daily MACD > Signal
- **Entry (H1):** When daily trend is bullish *and* H1 MACD turns bullish
- **Exit:** The strategy holds the long position for exactly one H1 bar after entry.
- **Rationale:** Higher timeframe reduces noise and avoids trades in consolidations

3.3 Trailing Exit Variant (STOP)

Same daily–hourly structure, but exit is improved:

- **Entry:** As in D1–H1 MACD
- **Exit:** The position is held until the first opposing candle, meaning:

The long position is closed when the next H1 candle closes below the previous H1 candle's low
(i.e., the first bearish candle relative to the prior bar).
- **Rationale:** Capture strong trends more effectively

4 Data

4.1 Dataset Overview

Table below summarizes all datasets used:

- **Asset:** BTC/USDT
- **Exchange:** Gemini
- **Frequency:** Hourly + Daily
- **Period:** Dec 31, 2018 – Nov 10, 2025

5 Results

5.1 Benchmark: Bitcoin Buy & Hold

Buy-and-hold produces strong long-term growth but suffers from multiple >60% drawdowns.

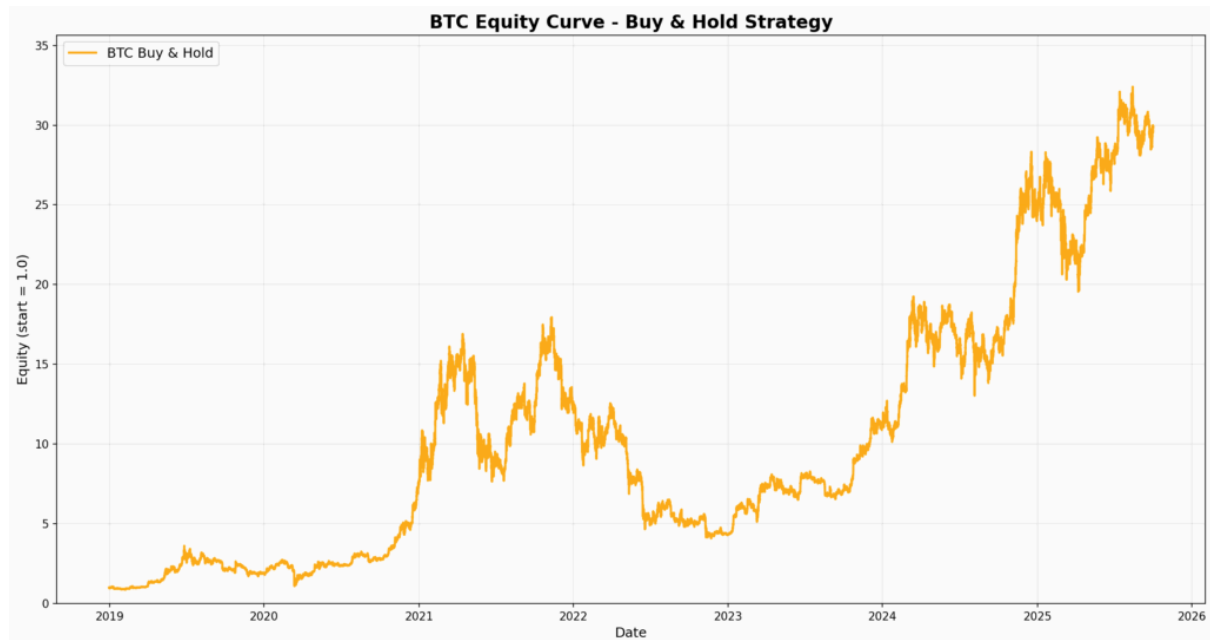


Figure 1: BTC Equity Curve — Buy & Hold Strategy

5.2 Baseline Strategy: MACD Pure (H1)

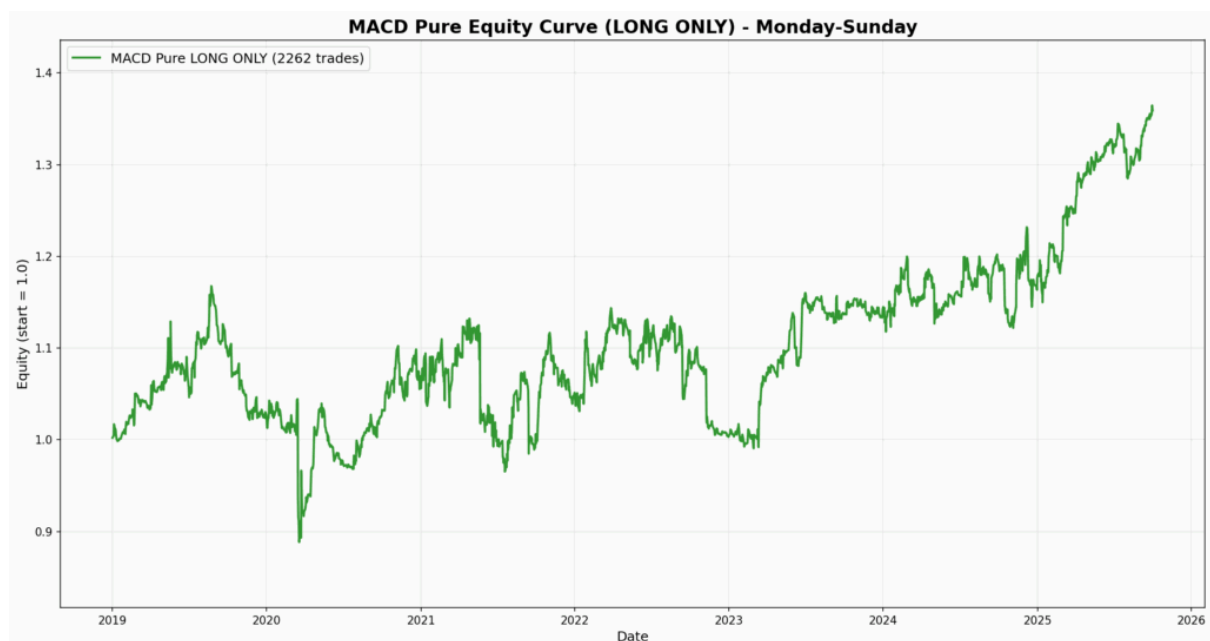


Figure 2: MACD Pure LONG ONLY Equity Curve

Metric	MACD Pure (LONG ONLY)
Number of Trades	2262
Annual Return (%)	4.61
Max Drawdown (%)	-23.91
Annual Volatility (%)	13.88
Sharpe Ratio	0.33
Calmar Ratio	0.19

Table 1: MACD Pure Performance Metrics

Interpretation:

- Weak annual return (+4.61%)
- Sharpe ratio 0.33
- Significant noise and whipsaws

This confirms that raw MACD is unsuitable for Bitcoin without additional structure.

5.3 Multi-Timeframe Strategy: D1–H1 MACD (Hold 1 Bar)

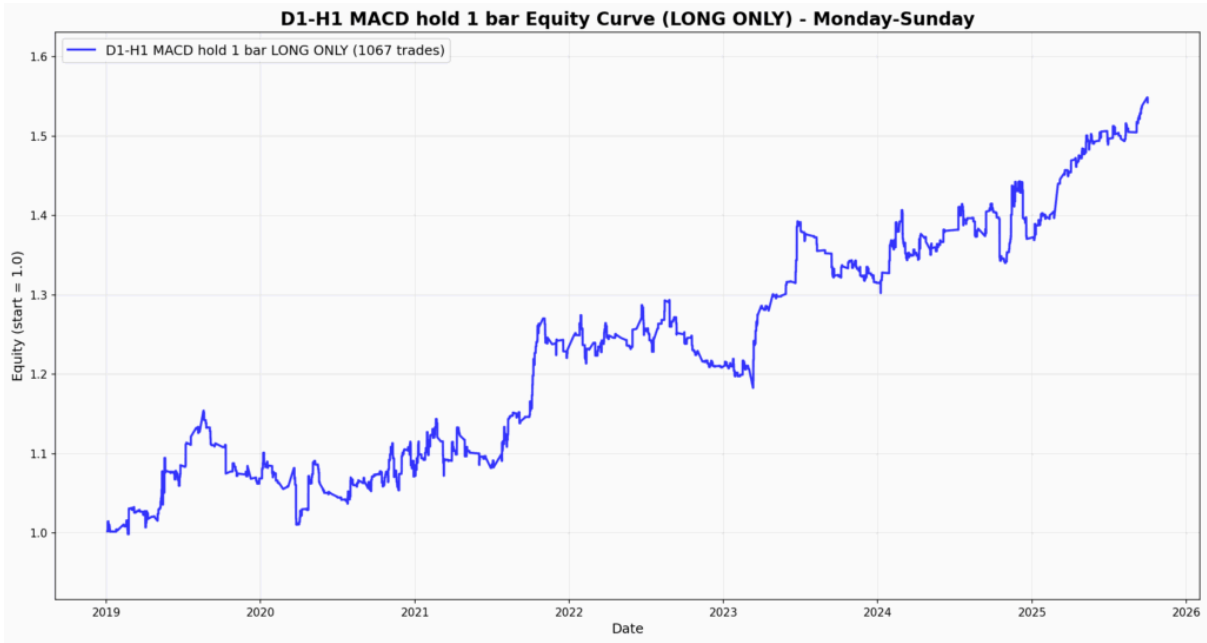


Figure 3: D1–H1 MACD hold 1 bar Equity Curve

Metric	D1-H1 MACD hold 1 bar (LONG ONLY)
Number of Trades	1067
Annual Return (%)	6.60
Max Drawdown (%)	-12.44
Annual Volatility (%)	8.26
Sharpe Ratio	0.80
Calmar Ratio	0.53

Table 2: D1–H1 MACD Performance Metrics

Interpretation:

- Annual return increases to 6.60%
- Sharpe ratio improves to 0.80
- Max drawdown decreases to -12.44%

Multi-timeframe filtering clearly improves trend reliability.

5.4 Trailing Exit Enhancement: D1–H1 STOP Strategy

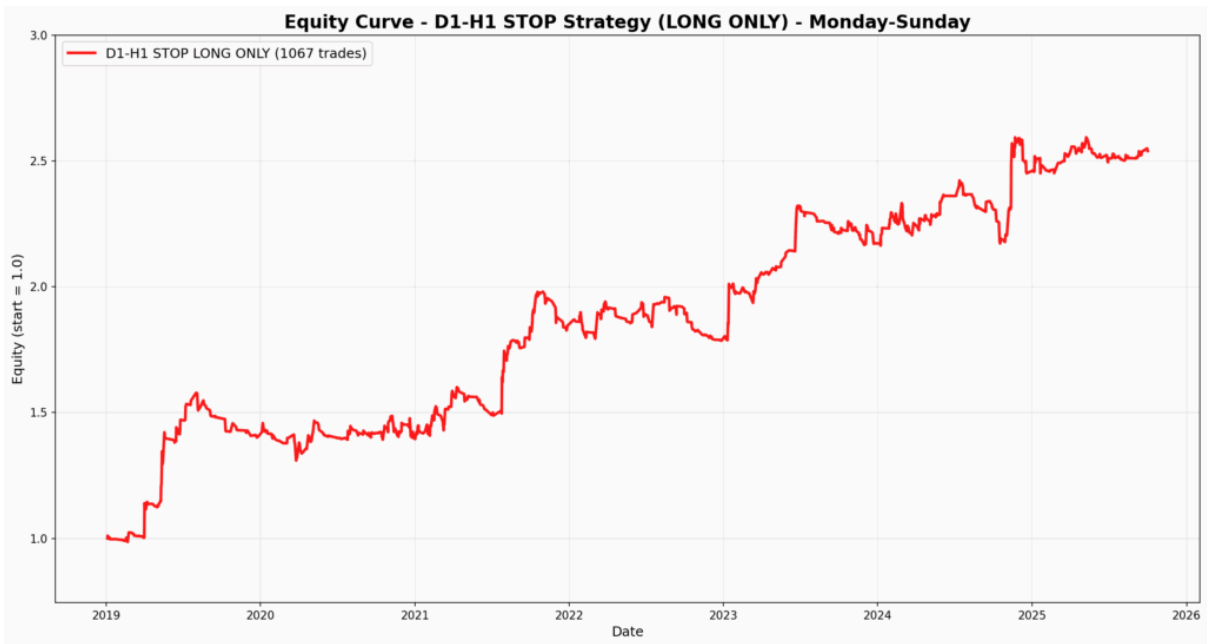


Figure 4: D1–H1 STOP Strategy Equity Curve

Metric	D1-H1 STOP (LONG ONLY)
Number of Trades	1067
Annual Return (%)	14.82
Max Drawdown (%)	-17.05
Annual Volatility (%)	13.80
Sharpe Ratio	1.07
Calmar Ratio	0.87

Table 3: D1–H1 STOP Performance Metrics

Interpretation:

- Annual return jumps to 14.82%
- Sharpe ratio increases to **1.07**
- Calmar ratio improves to **0.87**
- Drawdowns remain far smaller than buy-and-hold

This confirms that the trailing-exit variant meaningfully enhances trend capture.

6 Conclusions

This study demonstrates a structured process for developing a robust Bitcoin trend-following strategy using multi-timeframe logic. Starting from a weak baseline MACD model, we introduced:

1. **Daily trend confirmation** → reducing noise
2. **Hourly execution** → improving responsiveness
3. **Trailing exit logic** → capturing large directional moves

Each step resulted in clear and measurable improvement:

- Sharpe ratio increased from **0.33** → **0.80** → **1.07**
- Drawdowns decreased substantially
- Equity growth became smoother and more stable

The findings confirm that robustness emerges from logical structure, not optimization. Multi-timeframe trend-following is a highly effective framework for Bitcoin and can be extended with volatility filters, regime classification, and dynamic sizing in future research.

Acknowledgments

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References

Elder, Alexander (2002). *Come Into My Trading Room: A Complete Guide to Trading*.

Supplementary Material

Additional charts and code are available on request.